Calculations report: TEST - verification

Job number: TEST-001

Customer: TEST

Revision A, 2026-10-23

Approved by TEST APPROVER (client-supplied)

The design rules are this software's defaults; the company has not confirmed them.

Basis of calculation

- Design current Ib = P / (k U cos phi); breaker In >= Ib, cable Iz >= In (IEC 60364-4-43 §433.1; ABB Electrical Installation Handbook Vol. 2, §2.1-2.3).
- Current-carrying capacity from IEC 60364-5-52 Annex B, corrected for ambient temperature and grouping.
- Voltage drop from the origin, from Schneider EIG Fig. G28 (copper) and ABB §2.2.2 (aluminium), against IEC 60364-5-52 Annex G or the company's limits.
- Fault level at a sub-board through its feeder, IEC 60909-0 (cmax 1.10); breaking capacity at or above it.
- Earth fault disconnection, IEC 60364-4-41 §411.4.4: Zs x Ia <= 0.95 U0, Ia the top of the breaker's instantaneous band (IEC 60898-1).
- Short circuit along the cable, ABB §2.4: the far-end Ikmin trips the breaker at once; k^2S^2 from Table 1 for comparison with the breaker's let-through I^2t.

Board DB-HALL

400 V, 3-phase, TN-S · Ik - kA · Ze - ohm · Incomer C63 4P

Circuit	Ib A	Protection	Cable	L m	dU %	Zs ohm	Zs max	k^2S^2 $(kA)^2s$
Hall lighting row 1 (TEST VALUE)	5.8	C10 1P	3G1.5 Cu PVC	30	2.25	-	2.194	0.03
Hall lighting row 2 (TEST VALUE)	5.8	C10 1P	3G1.5 Cu PVC	35	2.62	-	2.194	0.03
Sockets hall east (TEST VALUE)	9.6	C16 1P	3G2.5 Cu PVC	25	1.87	-	1.372	0.083
AC unit (TEST VALUE)	8	C16 3P	5G2.5 Cu PVC	20	0.6	-	1.372	0.083

Notes

- Group 1 breaker raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination.
- Phase imbalance 21.8 % exceeds the company's 10 %: the loads cannot be spread more evenly as given.
- Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.
- No fault level given: every breaker's breaking capacity is left to be confirmed.
- Leave 1 spare outgoing ways (20 %).

- Loads without a power factor were taken at cos phi 0.9.

- Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.

- Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.

- Circuit rules are this software's defaults, not confirmed by the company's engineers.

- Enclosure: ABB Mini Center compact GDMS332 RX (2CVA250001P0001), 2 rows of 16 modules, H x W x D 498 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 15 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted.

Prepared	Checked	Approved
V4	TEST CHECKER	TEST APPROVER (client-supplied)